

in a system. While much work for this feature must be done in software, there are hardware features which need to be in place for a more robust design.

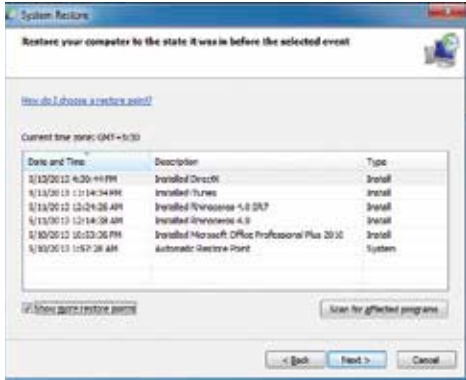
Robustness, at the hardware level, requires techniques to ‘isolate faults’. Since all components on the system consume the processing power of the same CPU, it can be difficult (and less effective) to design a system whose fault detection system is located centrally and takes care of all components. On the other hand, if each component comes with its own fault detection system, it simplifies the design at many levels and offloads ‘fault detection’ from the main processing system thus improving both, performance and robustness.

There are two concepts you need to understand for better fault isolation:

- 1 Temporal Partitioning: It means that one component cannot affect the guaranteed availability of communication resources available to another component. Here, by communication resources, we mean resources such as time of availability, latency, noise etc. Imagine your phone stops responding while trying to connect to a Wi-Fi hotspot because the Wi-Fi adapter clogged resources available to other parts of your phone. Temporal partitioning avoids this.

- 2 Spatial Partitioning: This not only improves robustness but also improves security. Spatial partition design ensures that the data integrity is maintained during communication between two components and that a third component cannot manipulate (or in some cases, even read) that data. This decoupled design ensures that if for some reason, one component is compromised, it won’t allow data to be stolen from other data streams. In other words, if your phone’s Bluetooth were to be hacked, the data you’re transferring via 3G network would not be available to the hacker/attacker.

Once again, both these features need a plethora of support from the software but the hardware design must support them as much as possible. Although, we’ve cited the example of devices used in our day-to-day activities to keep it as simple to understand as possible, there are many other embedded systems where these design features are highly anticipated.



Windows System Restore saves the states of the OS at different times